

Suno–CALM: A Dual-Track Proposal for Affective Robotic Dance via Semantic Audio Features with Latent RL and Diffusion Policies

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Abstract—We present Suno–CALM, a dual-track, modular research program for affective robotic dance that unifies: (i) a *physics-grounded latent RL controller* (CALM) and (ii) an *audio-conditioned diffusion policy*. Both tracks are grounded in semantic audio representations—centered on multi-scale embeddings from MERT and CLAP—explicitly used as *feature values* (beat/phase, energy, pitch/prosody, timbre/brightness, style/genre, and affect). A kinematic *adaptor* trained with InfoNCE aligns audio latents with a motion manifold learned from AIST++ and FineDance, enabling beat-precise, emotionally coherent, long-horizon motion generation in GPU-accelerated simulators (e.g., Isaac Gym, Isaac Lab). The diffusion track further explores retargeting-free, content-style disentanglement and teacher–student distillation with fast DDIM sampling for low-latency control (inspired by RoboPerform [1]), while following a staged path from strong EDGE-style [2] baselines toward beat-aware conditioning, lighter music encoders, and more sequence-efficient denoisers. In the current implementation stage, we focus on Track VI. We have established an AudioMimic–SONIC prototype in which an EDGE-style diffusion model trained on G1-retargeted AIST++ motions generates G1-compatible dance references from Jukebox music features and BeatDistance conditioning. The generated references can be replayed offline or streamed online through ZMQ to a SONIC whole-body controller for MuJoCo Sim2Sim execution. This current prototype provides the experimental foundation for subsequent semantic audio extensions with Librosa, CLAP, and MERT features.

I. INTRODUCTION

Robotic dance typically attains beat tracking but struggles with expressive, affect-aligned motion and long-horizon stability. Data scarcity in paired (music, emotion, 3D motion) and fragile whole-body control are key obstacles. We propose a *decoupled* approach:

- 1) **Affective Intent:** Suno AI provides diverse, controllable music to define emotional goals.
- 2) **Semantic Audio Features:** MERT [3] and CLAP [4] extract structured feature values from raw audio.
- 3) **Physical Execution:** Two complementary controllers: a physics-aware CALM (latent RL) and an *audio-conditioned diffusion* policy.

a) Why maintain two tracks?: Track A (CALM) emphasizes physical feasibility and long-horizon stability using a learned motion latent space and adversarial realism [5]. Track B (Diffusion) follows the retargeting-free, content-style paradigm with teacher–student distillation and audio-as-style injection to achieve expressive freestyle with low latency [1]. Within Track B, we explicitly follow a staged diffusion agenda: begin from strong diffusion baselines such

as EDGE [2] with Jukebox conditioning [6], introduce explicit beat modelling inspired by Beat-It [7], and later investigate lighter music encoders such as MERT [3] and more sequence-efficient backbones such as Mamba [8]. **Both tracks** reuse the same evaluation protocol and a largely shared semantic interface, enabling fair comparison and shared ablations.

b) Current implementation status.: Although Suno–CALM is formulated as a dual-track research program, our current implementation focuses on Track VI, the audio-conditioned diffusion path. We have implemented an EDGE-style AudioMimic baseline trained in the Unitree G1 motion space. Specifically, AIST++ dance motions are first retargeted to G1 using a GMR-based pipeline, and the diffusion model is trained to generate G1-compatible whole-body reference motions rather than SMPL-space motions.

The current checkpoint uses **Jukebox** music features together with explicit **BeatDistance** conditioning. At inference time, custom music can be processed into Jukebox features and audio-derived beat tracks, and the diffusion model generates G1 reference motions either offline as `.pkl` files or online through chunk-wise DDIM streaming. These references are transmitted through ZMQ to a SONIC whole-body control policy and executed in MuJoCo simulation. Therefore, the present system has already reached a working *music-to-G1-reference-to-SONIC-execution* prototype, while MERT, CLAP, richer Librosa features, and InfoNCE-based audio–motion alignment remain planned extensions.

II. DATA RESOURCES AND RETARGETING

Suno AI supplies controllable prompts for high- and low-arousal regimes (e.g., aggressive EDM vs. ambient music). **AIST++** [9] and **FineDance** [10] serve as the primary music–motion resources for learning dance priors.

a) Current implementation: G1-native diffusion training.: In the current AudioMimic implementation, we focus on AIST++ as the first-stage training source. Rather than training the diffusion model in the original SMPL/SMPL-H human motion space and retargeting generated motions at inference time, we first retarget the AIST++ motions to the Unitree G1 humanoid using a GMR-based retargeting pipeline. This produces a G1-space motion dataset containing robot-compatible joint trajectories and root states.

The diffusion model is then trained directly on these retargeted G1 motions:

$$\begin{aligned} &\text{AIST++ human motion} \\ &\xrightarrow{\text{GMR retargeting}} \text{G1 reference motion dataset} \quad (1) \\ &\xrightarrow{\text{diffusion training}} \text{AudioMimic model.} \end{aligned}$$

At inference time, the trained model therefore directly generates G1-compatible whole-body reference motions:

$$\text{music features} + \text{beat condition} \rightarrow \text{diffusion model} \rightarrow \text{G1 reference motion}$$

These generated G1 references can be streamed to the SONIC whole-body controller for MuJoCo Sim2Sim execution without requiring an additional online SMPL-to-G1 retargeting stage. We refer to this setting as *inference-time retargeting-free*: retargeting is still used offline during dataset construction, but the deployed generator operates directly in the robot motion space.

b) Retargeting as a future comparison axis.: Although GMR is used in the current implementation, we retain retargeting as an explicit research axis rather than a fixed design choice. Future experiments may compare GMR with alternative retargeting pipelines, including intermediate-skeleton retargeting, PHC-style retargeting, or other motion-optimization-based methods. This comparison is important because the quality of the retargeted G1 dataset directly affects diffusion training: abrupt transitions, joint spikes, root discontinuities, or unstable contact patterns in the retargeted data can be inherited by the generative model and later amplified during SONIC execution.

c) Extension to additional datasets.: While AIST++ is used as the current first-stage dataset, FineDance remains an important future extension because it provides more fine-grained choreography and stylistic diversity. A natural next step is therefore:

$$\begin{aligned} &\text{music features} + \text{beat condition} \\ &\xrightarrow{\text{AudioMimic diffusion}} \text{G1 reference motion} \quad (2) \\ &\xrightarrow{\text{ZMQ streaming}} \text{SONIC execution.} \end{aligned}$$

This allows us to evaluate whether additional dance diversity improves style coverage, motion realism, beat alignment, and robustness under whole-body execution.

III. AUDIO REPRESENTATION (MERT/CLAP AS FEATURE EXTRACTORS)

A. Feature values we explicitly extract

We compute a fused audio latent $l_{\text{audio}} = [z_t; z_s; \phi]$ with the following semantics:

- **Rhythm / Beat & Phase**: tempo/onset strength and normalized beat phase $\phi = \text{mod}(t, T_{\text{beat}})/T_{\text{beat}}$ for hit-the-beat precision.
- **Energy / Dynamics**: short-time energy or loudness proxies to scale motion amplitude and arousal.
- **Pitch / Prosody**: fundamental frequency contour and emphasis to modulate verticality and gesture emphasis.

- **Timbre / Brightness**: spectral centroid/flatness informing sharpness vs. smoothness in motion texture.
- **Style / Genre**: global stylistic cues (jazz, folk, EDM, etc.) to set dance texture.
- **Affect Signals**: coarse arousal/valence proxies to steer motion mood.

Here, MERT [3] provides multi-layer acoustic & temporal cues (z_t); CLAP [4] provides global semantic/style/affect cues (z_s); ϕ anchors beat-relative time. While this shared semantic front-end is centered on MERT/CLAP, Track VI will also benchmark Jukebox [6], following EDGE [2], as a deliberately strong diffusion-oriented music baseline.

B. Audio–Motion Alignment (InfoNCE Adaptor)

We learn a kinematic *adaptor* that projects the fused audio latent $l_{\text{audio}} = [z_t; z_s; \phi]$ into the motion latent space. The goal is to make audio descriptors act as *navigable directions* for control, rather than generic conditioning noise, so both CALM and Diffusion controllers can query audio as a semantically aligned control signal.

a) Objective (InfoNCE).: Given a minibatch of N paired audio–motion latents $\{(l_{\text{audio}}^{(i)}, l_{\text{motion}}^{(i)})\}_{i=1}^N$, where $l_{\text{motion}}^{(i)}$ is obtained from the motion encoder (on AIST++/FineDance), we treat (i, i) as the positive pair and all (i, j) , $j \neq i$ as negatives. Let $\text{sim}(u, v) = \frac{u^\top v}{\|u\| \|v\|}$ denote cosine similarity, and $\tau > 0$ be the temperature. The InfoNCE loss is

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(\mathcal{A}(l_{\text{audio}}^{(i)}), l_{\text{motion}}^{(i)})/\tau)}{\sum_{j=1}^N \exp(\text{sim}(\mathcal{A}(l_{\text{audio}}^{(i)}), l_{\text{motion}}^{(j)})/\tau)}, \quad (3)$$

where $\mathcal{A}(\cdot)$ is the adaptor network that maps audio latents into the motion-latent space.

b) Why it works (alignment & uniformity).: The numerator in (3) pushes the adapted audio latent $\mathcal{A}(l_{\text{audio}}^{(i)})$ to be close to its matched motion latent $l_{\text{motion}}^{(i)}$ (*alignment*). The denominator (with all negatives) prevents collapse by spreading samples over the manifold (*uniformity*), so the adaptor learns a geometry where audio cues (beat, energy, style/affect) correspond to meaningful directions in the motion space. The temperature τ controls the sharpness of the contrast: smaller τ yields peakier softmax (harder negatives), while larger τ leads to smoother gradients.

c) Positive/negative construction.: We use synchronized audio–motion clips as positives. All other motion latents within the same batch form in-batch negatives (no extra memory bank is required). Empirically, larger and more diverse batches improve the quality of negatives and the robustness of the learned alignment.

d) Implementation details.: (1) We ℓ_2 -normalize both $\mathcal{A}(l_{\text{audio}}^{(i)})$ and $l_{\text{motion}}^{(i)}$ before similarity; (2) the adaptor is a lightweight temporal module (self-attention / temporal Conv/MLP) to absorb rhythmic structure; (3) we optionally add a small *alignment regularizer* $\|\mathcal{A}(l_{\text{audio}}) - \text{sg}(l_{\text{motion}})\|_2^2$ with stop-gradient on motion latents to stabilize early training; (4) the adaptor output dimensionality matches the motion-latent dimension.

IV. SIMULATION AND TRAINING

We support multiple GPU-accelerated simulators. Unless otherwise specified, experiments are run in one of: *NVIDIA Isaac Gym* (link), *NVIDIA Isaac Lab* (link), or an equivalent physics engine (e.g., MuJoCo), chosen per hardware and licensing constraints. Our code abstracts the simulator backend behind a unified interface (sensors, contacts, action rate), enabling fair comparison across controllers and seamless backend switching for ablations and deployment preparation.

V. TRACK A: CALM (CONDITIONAL ADVERSARIAL LATENT MODELS)

A. Latent Motion Manifold

We learn an encoder-decoder E/D over dance clips with an adversarial discriminator enforcing realism on AIST++ and FineDance, yielding a compact motion latent c . The design follows the spirit of conditional adversarial latent modeling for directable characters [5].

B. Semantic-Conditioned Latent RL

A policy outputs target latent codes based on robot state and audio features:

$$c_t \sim \pi(s_t, z_t, z_s, \phi), \quad q_t = D(c_t).$$

The latent interface confines exploration to human-like manifolds, stabilizing long-horizon control.

C. Training in GPU Simulators and Reward

Training uses massively parallel GPU simulators (Isaac Gym / Isaac Lab / MuJoCo) with domain randomization. The reward

$$R = w_a R_{\text{adv}} + w_s R_{\text{sync}} + w_p R_{\text{pen}}$$

combines adversarial realism, beat alignment (e.g., $\exp(-\alpha|v_{\text{foot}} - v_{\text{beat}}|)$), and penalties (torque limits, self-collision).

VI. TRACK B: AUDIO-CONDITIONED DIFFUSION POLICY

A. Motivation and high-level design

We adopt a *retargeting-free* diffusion policy that treats **motion = content + style**: a *content latent* l_{content} (from a text-to-motion prior defining “dancing/gesturing”) preserves task semantics; the aligned *audio latent* l_{audio} injects style (rhythm/prosody/energy). This design is amenable to *teacher-student distillation* and *fast DDIM sampling* for low-latency control, as suggested by RoboPerform [1]. As the main diffusion backbone, we take EDGE [2] as the conceptual starting point: it demonstrates that transformer-based diffusion with a strong music encoder such as Jukebox [6] can produce realistic, editable, and long-horizon dance sequences. We then extend this line with explicit beat modelling, stronger semantic conditioning, and later architectural upgrades.

B. Beat-aware diffusion conditioning

Beyond global semantic/style conditioning, we further equip the diffusion track with an explicit beat-aware control pathway, inspired by the beat-specific conditioning philosophy of Beat-It [7]. The motivation is that general audio embeddings provide broad musical context, whereas structured beat representations provide precise temporal cues for rhythmic synchronization. Let h_{music} denote the general music condition and let h_{beat} denote an explicit beat representation derived from the same waveform. The diffusion policy is conditioned on both:

$$\hat{x}_0 = f_{\theta}(x_t, t, l_{\text{content}}, h_{\text{music}}, h_{\text{beat}}),$$

where x_t is the noisy motion sample at diffusion step t , and $\hat{x}_0$ is the denoised motion prediction. In the staged development of Track VI, we consider two beat representations:

- **BeatPulse**: a binary beat pulse / beat mask aligned to the motion timeline.
- **BeatDistance**: a nearest-beat-distance representation that encodes each frame’s relative position to the nearest beat.

A corresponding beat-aware objective may be added in later stages:

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda_{\text{beat}} \mathcal{L}_{\text{beat}},$$

where $\mathcal{L}_{\text{diff}}$ is the original diffusion loss and $\mathcal{L}_{\text{beat}}$ encourages synchronization between motion accents and music beats. This design lets us separate the contribution of explicit beat input from the contribution of beat-specific supervision.

C. Current AudioMimic-SONIC Prototype

The current prototype implements the first stage of Track VI: a beat-conditioned EDGE-style diffusion baseline connected to a whole-body execution stack. The model checkpoint currently used in our experiments is trained under the `g1_aist_beatdistance_featurecache` setting. It consumes Jukebox music features and BeatDistance beat representations:

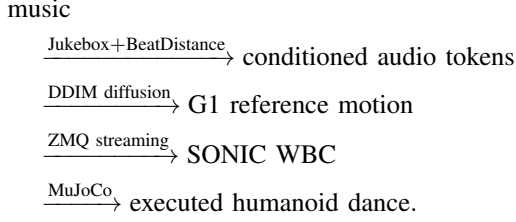
$$h_{\text{music}} \in \mathbb{R}^{T \times 4800}, \quad h_{\text{beat}} \in \mathbb{Z}^T,$$

where $T = 150$ for a 5-second window at 30 FPS. The BeatDistance representation encodes the distance from each frame to its nearest beat, providing a dense temporal scaffold that is more informative than a sparse BeatPulse mask.

At inference time, the system supports two modes. In the offline mode, AudioMimic first generates a G1 motion file, which is then replayed through a ZMQ streamer. In the online mode, DDIM inference is performed chunk-by-chunk and streamed directly to SONIC. Each diffusion window spans 5 seconds, with a 2.5-second stride and a 2.5-second overlap. The overlap region is blended by crossfade interpolation to reduce discontinuities between chunks.

The generated G1 references are transmitted through ZMQ to the SONIC C++ deployment binary, which tracks the incoming reference using its whole-body control policy in Mu-

JoCo simulation. This gives us a practical Sim2Sim execution loop:



The current checkpoint is only compatible with Jukebox features and BeatDistance conditioning. Future semantic audio extensions with MERT, CLAP, and additional Librosa-derived features will therefore be introduced through compatible gated fusion or fine-tuning rather than direct replacement of the current conditioning interface.

D. Phased milestones for Track B

We envision Track VI as a staged program with progressively richer conditioning, stronger music representations, and more efficient sequence backbones.

a) Milestone 1: Beat-conditioned EDGE-style baseline.:

We begin with an EDGE-style diffusion baseline [2] augmented with explicit beat representations. Here, Jukebox [6] acts as a deliberately strong music baseline, allowing us to test whether structured beat cues improve beat alignment and rhythmic controllability beyond general music conditioning alone.

b) Milestone 2: Audio encoder upgrade and comparison.:

We then compare alternative music encoders within the same beat-aware diffusion framework. In particular, MERT [3] becomes a promising upgrade candidate because it is designed for music understanding rather than raw-audio generation. This stage focuses on the trade-off between motion quality, beat synchronization, latency, and compute cost when moving from a strong but heavy Jukebox baseline toward lighter semantic audio encoders.

c) Milestone 3: Backbone evolution beyond transformers.:

Finally, we investigate whether the transformer denoiser can be replaced or augmented by more sequence-efficient architectures. A strong motivation comes from Mamba [8], which offers linear-time sequence modeling and hardware-aware efficient inference; these properties are particularly attractive for long audio–motion horizons where quadratic attention can become a bottleneck. Recent dance-generation systems such as Danceba [11], MambaDance [12], and LRCM [13] suggest that Mamba-style modules are promising for rhythm-aware and long-horizon dance synthesis. In our setting, the goal is not to assume that Mamba must replace transformers outright, but to treat Mamba-style denoisers as a principled avenue for improving long-horizon coherence, scaling behavior, and deployment efficiency.

E. Conditioning and inference

Audio is injected across diffusion layers (e.g., AdaLN) to steer denoising toward rhythmically stylized and physically executable actions. In early variants, this conditioning

combines general music embeddings with an explicit beat branch; later variants extend the same interface to stronger semantic front-ends and more efficient denoisers. A 2–4 step DDIM schedule targets sub-10 ms policy latency for real-time deployment, reusing the same adaptor abstraction and beat-aware conditioning API across staged versions of Track VI.

VII. EVALUATION PLAN (SHARED BY BOTH TRACKS)

A. Core Metrics

- 1) **Beat Alignment Score (BAS):** kinematic vs. music beats (beats from Librosa or equivalent onset detectors).
- 2) **Fréchet Dance Distance (FDD):** distributional realism vs. real dances.
- 3) **Affective Consistency:** intended vs. perceived affect.
- 4) **Long-Horizon Stability:** uninterrupted duration without falls or rhythm loss.
- 5) **Latency (Diffusion):** per-step wall-clock time under target hardware.

B. Current Execution-Aware Metrics

For the current AudioMimic–SONIC prototype, we additionally evaluate whether the generated references remain physically executable after whole-body tracking:

- 1) **Reference tracking error:** RMSE between the streamed G1 reference joint positions and the executed joint positions under SONIC.
- 2) **Stability:** fall rate, survival duration, root height, and base roll/pitch statistics in MuJoCo.
- 3) **Joint safety:** joint limit violations, velocity spikes, and acceleration discontinuities in generated and executed motions.
- 4) **BAS_{ref}:** Beat Alignment Score computed on the generated G1 reference motion.
- 5) **BAS_{executed}:** Beat Alignment Score recomputed from the physically executed trajectory after SONIC tracking.
- 6) **Latency:** Jukebox feature extraction time, DDIM generation time, ZMQ streaming delay, and SONIC control-loop latency.

C. Key Ablations

- **Front-end:** MERT-only / CLAP-only / fused; w/ vs. w/o ϕ ; w/ vs. w/o InfoNCE adaptor.
- **Controller:** CALM vs. Diffusion; content–style vs. content-only; DDIM steps vs. latency.
- **Beat-aware conditioning (Track B):** no explicit beat conditioning / BeatPulse / BeatDistance / BeatDistance+ $\mathcal{L}_{\text{beat}}$.
- **Track-B architecture evolution:** Jukebox vs. MERT front-end; transformer vs. Mamba-style backbone; compare beat alignment, motion realism, diversity, long-horizon stability, and latency.
- **Stability:** latent prior on/off; adversarial term on/off; torque/contacts penalties scaling.

VIII. PROGRAM MODULARIZATION (2–3 STUDENTS)

a) *Student A: Audio–Motion Representation.:* Suno → MERT / CLAP → adaptor; implement and benchmark BAS/retrieval; ship a stable l_{audio} API for both tracks.

b) *Student B: Track A (CALM).*: Train latent RL with adversarial prior and beat-phase conditioning in a GPU simulator backend (Isaac Gym / Isaac Lab / MuJoCo); target long-horizon stability; report BAS/FDD/energy.

c) *Student C: Track B (Diffusion).*: Implement the staged diffusion agenda, beginning with an EDGE-style beat-conditioned diffusion baseline, beat-aware ablations, and Jukebox/MERT front-end comparisons; then extend toward content-style conditioning, AdaLN-based injection, DDIM 2–4 step sampling, and sequence-backbone upgrades such as Mamba-style denoisers, together with latency/stability trade-off analysis (cf. RoboPerform [1]).

IX. CONCLUSION

Suno–CALM offers a dual-track, modular path to affective robotic dance: a physics-grounded CALM controller and an audio-conditioned diffusion policy, *both* powered by explicit music feature values extracted via MERT/CLAP and aligned to motion latents via InfoNCE. Within this broader agenda, Track VI is developed as a staged line of research: from a strong EDGE-style/Jukebox baseline, to beat-aware conditioning, to later encoder comparisons with MERT, and eventually to Mamba-style sequence backbones for improved long-horizon scalability and runtime efficiency.

At the current implementation stage, we focus on Track VI. We have established an AudioMimic–SONIC prototype in which AIST++ dance motions are first retargeted to the Unitree G1 using a GMR-based pipeline, and the resulting G1-space motions are used to train an EDGE-style diffusion model. The current checkpoint uses Jukebox music features and BeatDistance conditioning to generate G1-compatible reference motions. These references can be produced offline as .pkl files or generated online through chunk-wise DDIM streaming, and are transmitted via ZMQ to a SONIC whole-body controller for MuJoCo Sim2Sim execution. This makes the current diffusion generator *inference-time retargeting-free*: retargeting is used offline during dataset construction, but the deployed model directly outputs robot-space G1 references.

The immediate next step is to evaluate this prototype under execution-aware metrics, including reference-level beat alignment, executed beat alignment after SONIC tracking, tracking error, stability, joint safety, and latency. In parallel, we will extend the current Jukebox+BeatDistance baseline with richer Librosa-derived acoustic features, CLAP global style/affect embeddings, and MERT temporal music representations through gated fusion. The shared front-end abstractions and unified evaluation protocol enable parallel progress and fair comparison across tracks.

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